

mvn_ngi_filelvlcng_v02_r01.pdf

Bold and starred level indicates most recent and current version to be used.

Level	Ver/Rev	Features	Major Changes
L1a	v01_r01**	Raw data of all orbits un calibrated (hsk, sci, msg etc) No alteration made from sc. All other products derived from this product.	This Product is never changed
L1b(sci)	v09_v01**	Calibrated data (osion, osnb, obion, wind**) Includes ephemeris and Background subtractions **wind in production not yet available	Corrected t_unix time drift from t_utc and t_sclk. This column slipped due to error in early computation step and kernel sclk update fault. Kernel update error has been corrected and t_unix computation has been updated with check to alert team for drift before delivery.
L2 ion	v08_r01**	Derived ion data product: contains all ion densities m2-150.	No substantial changes from version 7 other than t_unix column updated along with l1b files. Flags simplified to only contain SC0, SCP, or SCH related to the spacecraft potential (none, present, or high)
L2 csn	v08_r01**	Derived neutral densities	Updated and corrected CO densities (corrected sensitivity, adjusted factor of 2 fixed mass error). Corrected N2 mass column to reflect mass of N2 rather than mass channel used (now reads m28). Updated T_unix column updated as affected by l1b. Ar densities extended to highest possible altitudes to aid new high-altitude measurements.

			flags (IV, OV-inbound & outbound verified; IU, OU inbound & outbound uncertain; UR unreliable, not recommended to use due to lack of background on instrumentation effect)
L3 sht	v06_r01**	Derived scale Height : scale heights & temperatures	Adjusted scale height altitude ranges slightly higher to more closely center around the isothermic range. Lower limit now defined either CO2densIN >1e8 or computed exo*0.8 whichever is higher and still give reasonable range*. Added error for computed scale height and derived temperature. (*NOTE* Columns have changed!) Flags: HI-good fit, near the isothermal range ($r^2 > 75\%$), HW, fit is to wavy or structured atmosphere, poor fit ($r^2 < 0.75$)
L3 den	v06_r01**	Derived densities: averaged over set bin	Previous versions the densities have been averaged over set altitude bins, this was determined to be skewed to the lower altitudes when the spacecraft is moving more horizontal. This version the bins are constant density by log(dens) in 0.2 binned steps. (*NOTE* columns have adjusted) the columns are slightly different to adjust for the different binning style. Flags still reflect l2 csf flags.
L1b (sci)	v08_r01	Calibrated data (osion, osnb, obion, wind**)	Added t_sclk_cor and t_sclk_raw and quality flags to distinguish corrections made

		Includes ephemeris and Background subtractions **wind in production not yet available	to ngims sclk due to differences between ngims sclk and sc sclk
L1b	v07_r02/03	Calibrated data (osion, osnb) Includes ephemeris and Background subtractions	Improved background subtraction method on neutral data (note not updated to v07 method as yet!!) corrected sclk drift (10/1/15-2/2/16)
L1b	v07_r01	Calibrated data (osion, osnb) Includes ephemeris and Background subtractions	add t_utc for ease of PDS search capacity
L1b	v06_r04	Calibrated data (osion, osnb) Includes ephemeris and Background subtractions	Background subtraction on neutral data uses first 50s of inbound dropping warmup for better accuracy. Checked ephemeris with updated kernels. Removed bad pointing l1b files (set as precedent for future delivery) Removed NaN to comply with PDS and corrected PDS label and floats to match PDS formatting
L1b	v06_r03	Calibrated data (osion, osnb) Includes ephemeris and Background subtractions	Background subtraction method same (r02) Updated ephemeris with updated kernels from naïf. Removed l1b files corresponding to COMM passes, unrecoverable data gaps or other bad pointing periapsis.
L1b	v06_r02	Calibrated data (osion, osnb) Includes ephemeris and Background subtractions	New background subtractions method r01 used linear fit to warmup, inbound and outbound, r02 averaged warmup and first 5 seconds of inbound
L1b	v06_r01	Calibrated data (osion, osnb) Includes ephemeris and Background subtractions	Altitude, Vsc and time formats were finally all agreed upon by all instrument teams in v06.

L1b	v05 – v01	Calibrated data (osion, osnb) Includes ephemeris and Background subtractions	v01-v05 most changes were in format with ephemeris discussions and background subtraction techniques. v04, v05 and v06 only versions published to SDC
L2 csn/cso	v07_r03 **	Derived neutral densities	Corrected CO densities to new method used for Ar, CO2, N2 and O. (was meant for r02 but missed due to computer error). Corrected output error in precision on all neutral species. Removed NO from CSO files due to uncertainty of instrumental effects.
L2 csn/cso	v07_r01 (cso/csn)	Derived neutral densities	Neutral only! Revised CO2, Ar, N2 and O neutral composition. For these species the background has a slightly different computation using both inbound and outbound (still in review) and has a revised method of computing overlap and pileup effects for periapsis. The largest change effects O, O2 and CO. No longer reporting CO from OS (redundant) removing O2 (OS) inaccurate and reporting O (cs) with the assumption that the m32 channel is from O recombining inside instrument for better accuracy
L2 ion	V07_r01 **	Derived ion densities	Added SC_potential (from STATIC) column
L2 ion	v06_r03	Derived ion densities	Revised ion density to confirm use of STATIC potential
L2 csn cso ion	v06_r02	Derived neutral and ion densities	Revised overlap and background computation for neutrals, updated ion sensitivity
L2 csn cso ion	v06_r01	Derived neutral and ion densities	Improved neutral computational method, including new quality flags

L2 all	v05_r02/03	Derived neutral and ion densities	Updated ion densities to STATIC/LPW, corrected sclk drift, adjusted neutral fractionation patterns
L2 all	v05_r01**	Derived neutral and ion densities	Added: t_utc, lat, long, lst, sza for ease of use. Updated neutral fractional calibrations (cal_gas 7/30/15). Updated ion calibrations with LPW (9/18/15). Updated error calculations (Poisson distribution & %error error should be plotted as $X = X \pm \%e$). Corrected errors from overlap calculations between attenuations.
L2 csn/cso	v04_r04**	Derived neutral data product: csn contains only closed source neutral product (Ar, CO2, He, N2) while cso contains data for all neutral products we can compute (Ar, CO2, He, N2, CO, O2, O, NO)	New adjustment method for overlap between attenuation cps for 44, 28, 40 and 16. Computing OS and CS** density for CO (cso only) for orbits greater than 714. Improved computation of O. [KP product will use CS value whenever possible]. Improved fractionation patterns and sensitivity for C, O, N, CO2, CO, He, and NO. Precision now computed as a percentage error. Interpolated all cps and density to 2s cadence closer to each point of measurement to reduce error in interpolations.
L2 csn/cso	v04_r03	Derived neutral data product: csn contains only closed source neutral product (Ar, CO2, He, N2) while cso contains data for all neutral products we compute (Ar, CO2, He, N2, CO, O2, O, NO)	Fixed error with csv column on O2 abundance/precision (period where a comma should be minor correction)
L2 csn/cso	v04_r02	Derived neutral data product: csn contains only closed source	**error in O2 abundance/column (period where comma should be)

		neutral product (Ar, CO ₂ , He, N ₂) while cso contains data for all neutral products we compute (Ar, CO ₂ , He, N ₂ , CO, O ₂ , O, NO)	combines the two columns in csv format. Added precision and quality columns. Precision in physical units of abundance (affected by variations in atmosphere). Updated fragmentation patterns. Interpolated to 1s cadence on cps and abundance to ease subtraction between interacting species (O, CO, N, C, etc.)
L2 csn/cso	v04_r01	Derived neutral data product: csn contains only closed source neutral product (Ar, CO ₂ , He, N ₂) while cso contains data for all neutral products we compute (Ar, CO ₂ , He, N ₂ , CO, O ₂ , O, NO)	Interpolated to 1s cadence. Updated fractionation patterns and corrected error on N ₂ sensitivity. All negative density values set to NaN to prevent confusion. Some errors at periapsis and in overlaps for CO, O and CO ₂ corrected in later formats.
L2 csn/cso	v03_r01	Derived neutral data product: csn contains only closed source neutral product (Ar, CO ₂ , He, N ₂) while cso contains data for all neutral products we can compute (Ar, CO ₂ , He, N ₂ , CO, O ₂ , O, NO)	No interpolations. This version generally worked through DD1 & 2 and was used to work through hi and lo emission changes.
L2 csn/cso	v02	Derived neutral data product: csn contains only closed source neutral product (Ar, CO ₂ , He, N ₂) while cso contains data for all neutral products we can compute (Ar, CO ₂ , He, N ₂ , CO, O ₂ , O, NO)	No time interpolations. Was used to determine count drift error from beginning of mission through February. There are many files from this version and v03 that are from December and January that were forced to 'work' in the code that do not have reliable results.
L2 csn/cso	v01	Derived neutral data product: csn contains only closed source neutral product (Ar, CO ₂ , He, N ₂) while cso	Generally first efforts to get siding spring and early mission results out. More calibration was done throughout mission.

		contains data for all neutral products we can compute (Ar, CO2, He, N2, CO, O2, O, NO)	
L2 ion	v04_r02**	Derived ion data product: contains all ion densities m2-150.	Interpolated to 2s cadence to match better with measurement times and with neutral computation. Improved overlap computation. Corrected skipping errors for overlap calculation for m32 corrections.
L2 ion	v04_r01	Derived ion data product: contains all ion densities m2-150.	Interpolated to 1s cadence, added precision and quality flags.
L2 ion	v03_r03	Derived ion data product: contains all ion densities m2-90.	Precision and quality flags, no interpolation. No roll over correction for m32, 16, 28 or 44.
L2 ion	v03_r02	Derived ion data product: contains all ion densities m2-90.	No precision/quality, no interpolation, no rollover correction. Adjusted sensitivity to LPW October and December data.
L2 ion	v03_r01	Derived ion data product: contains all ion densities m2-90.	Can account for masses up to 150 when script call for it, and sensitivity adjusted to LPW data October and December.
L2 ion	v01&v02	Derived ion data product: contains all ion densities m2-90.	Can only account for m2-90. Adjusted sensitivity only to LPW October data.
L3 sht (den no change from previous)	v05_r01 **	Derived averaged density product & scale Height : average density over 10 km steps; scale heights & temperatures	Updated scale heights to fit between computed exobase altitude and homeopause. Added columns for fit residuals, mean of fit and std of fit to assess quality of fit. Change avg_alt to computed exobase altitude. Exobase altitude based on method in M.slipski et al 2016.
L3 all	v04_r01	Derived averaged density product & scale Height : average density	Updated to reflect changes made in the neutral l2 computations.

		over 10 km steps; scale heights & temperatures	
L3 all	v03-v01	Derived averaged density product & scale Height : average density over 10 km steps; scale heights & temperatures	All previous versions were updated to reflect changes made in the l2. Each update to the L2 neutrals will increase a revision or version change in the l3.